Nedu Ekeruche

Toronto • Ontario
cekeruche@icloud.com - Linkedin

EDUCATION

Carleton University **Ottawa, ON**
Bachelors of Mathematics and Statistics [90% GPA] *September 2018 - June 2023*

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, Go, Typescript

Frontend: React, React Native, Expo

Data & Infrastructure: Airflow, Apache Spark, Pandas, NumPy, FastAPI, Redis, pgvector

Cloud & Orchestration: GCP(Vertex AI, Cloud Functions, Cloud Run), AWS, Docker, Terraform, GitHub Actions.

Databases: PostgreSQL, MongoDB, BigQuery

WORK EXPERIENCE

Software Developer
SevenDot Analytics *Jan 2024 - current*

- Developed and maintained scalable backend services in Python, optimizing scripts and data processing for real-time systems.
- Built and optimized complex PostgreSQL queries, improving response time by 25% for large-scale data ingestion and retrieval.
- Implemented backend APIs and WebSocket-based real-time data pipelines, reducing latency by 35% and supporting critical business decisions.
- Led testing efforts using Pytest, achieving 70% test coverage and increasing deployment confidence.
- Wrote custom scheduling and orchestration flows in Airflow to streamline data ingestion, transformation, and validation.
- Implemented mobile features using React Native and Expo based on provided UI designs, focusing on smooth data flow, API integration, and responsive user interactions to enhance overall app functionality.

Freelance Software Developer
ShutterUnit Photography *May 2024*

- Architected and deployed a production-grade Next.js + TypeScript media platform serving 10,000+ high-resolution assets with optimized image processing, lazy loading, and responsive design patterns.
- Engineered CDN integration with intelligent caching layers and asset optimization pipelines, reducing page load times by 40% and improving Core Web Vitals scores for global users.
- Implemented cloud-native infrastructure with auto-scaling capabilities, load balancing, and fault-tolerant architecture to handle traffic spikes with 99.9% uptime and zero-downtime deployments.

PROJECTS

- **Craftha Marketplace**, craftha.com - Architected and developed a production-grade full-stack e-commerce marketplace platform connecting artisans with customers, featuring separate web and mobile applications for buyers and sellers. Built scalable backend services using FastAPI, PostgreSQL with pgvector for semantic search, and Redis for caching, implementing Stripe Connect for marketplace payment processing with automated seller payouts. Developed responsive Next.js frontend with TypeScript and React Native mobile apps using Expo, delivering real-time messaging, order management, shipping profiles, analytics dashboards, and AI-powered product recommendations. Engineered comprehensive seller onboarding flows, inventory management, and admin tools with Docker containerization for deployment.
- **SereinMod** (sereinmod.com), Architected and deployed a production-grade AI-powered content moderation platform using Next.js, FastAPI, PostgreSQL, and Redis, serving real-time moderation decisions with sub-100ms latency. Built custom ML model serving infrastructure using scikit-learn with TF-IDF vectorization, detecting 6 toxicity categories with configurable thresholds. Implemented comprehensive analytics dashboards, webhook event system with retry logic, multi-project API key management, and async background task processing. Engineered Redis caching layers, rate limiting, and health monitoring systems to ensure 99.9% uptime for enterprise clients.

- **Whot Online**, Architected a full-stack real-time multiplayer card game using React, FastAPI, WebSockets, and Redis, implementing complex Nigerian Whot game rules with special card mechanics. Built a responsive React frontend with real-time state synchronization and designed async backend game state management with connection pooling, player reconnection handling, and spectator mode. Delivered scalable session management supporting concurrent games with sub-100ms latency for seamless gameplay synchronization across clients.

CERTIFICATIONS

- Google Professional Machine Learning Engineer Sept 2023
- Google TensorFlow Developer Sept 2023